

Amirhosein Amini

PHD STUDENT IN CHEMISTRY | PREFERRED NAME: AMIR

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Education

Texas A&M University

PhD Student in Chemistry (Advisor: Dr. Arkajit Mandal)

- Developing cavity QED electronic structure methods and applying them to coupled light-matter systems.

College Station, TX, USA

January 2025 – Present

Sharif (AryaMehr) University of Technology

BSc in Chemistry with a Minor in Physics (*summa cum laude*)

- Investigated computational spectroscopy methods and quantum dynamics under the supervision of Dr. Zahra Jamshidi.

Tehran, Iran

September 2019 – June 2024

Publications

† Equal Contribution, * Corresponding Author

[1] On-the-fly cavity-molecular dynamics of vibrational polaritons

S Wickramasinghe†, **A Amini**†, A Mandal*

Physical Chemistry Chemical Physics 2026

[2] Quantum Ninjas vs. Laser Robots: A Board Game to Cultivate Quantum Intuition

A Amini, P Ghosh, L Blackham, A Manjalingal, S Rahmanian Koshkaki, S Wickramasinghe, SM Lim, A Mandal*

ChemRxiv 2026

[3] Giant optical spin-orbit interactions in ferroelectric van der Waals waveguides

D Xu†, S Rahmanian Koshkaki†, V Galicia, CY Huang, V Quiros-Cordero, JA Tulyagankhodjaev, A Koch Liston, **A Amini**, Y Hong, T Handa, PJ Schuck, X Zhu, X Roy, A Mandal*, M Delor*

Submitted to Nature 2026

[4] Many-Body Second Order Green's Function Theory for Polaritons

A Amini, J Cerda, L Mejía, A Mandal*

In Preparation 2026

Achievements

2026 **Journal Cover Article** — *Physical Chemistry Chemical Physics*, Vol. 28, No. 13 (April 2026), Royal Society of Chemistry

2019 **Ranked among the Top 1% of more than 165,000 examinees in Iran's Universities' Nationwide Entrance Exam**, Iranian National University Entrance Exam

2024 **Award of Extracurricular Activities among B.Sc. Students of Chemistry Department**, Sharif University of Technology

2023 **Executive Committee Member, Winter School of Computational Chemistry**, Sharif University of Technology

Teaching

Texas A&M University

Teaching Assistant / Instructional Assistant

Spring 2025 – Present

- Teaching Assistant for CHEM 119: "Fundamentals of Chemistry I" (Spring 2025).
- Teaching Assistant for Physical Chemistry II Lab, Computational Chemistry module (Fall 2025).
- Instructional Assistant for Fundamentals of Chemistry for Engineering Students (Fall 2025).
- Teaching Assistant for Physical Chemistry II Lab, Computational Chemistry module (Spring 2026).
- Teaching Assistant for Physical Chemistry I Lab — Atomic Force Microscopy (AFM) module (Spring 2026).

Outreach and Professional Activities

Texas A&M University

College Station, TX, USA

Outreach, Service, and Software Contributions

2025 – Present

- Peer Reviewer, The Journal of Chemical Physics** — Reviewed manuscripts in computational and theoretical chemistry.
- cavOTF Software Development** — Contributed to the development of the **cavOTF** package for cavity molecular dynamics (GitHub).
- Physical Chemistry II Laboratory Curriculum Development** — Authored and developed the Computational Chemistry module (Experiment 1: Introduction to Quantum Chemistry).
- Poster Presentation, Texas Quantum Summit** — Presented research on cavity QED and quantum dynamics.
- Chemistry Open House Outreach** — Led interactive demonstrations introducing quantum mechanics concepts through the board game *Quantum Ninjas vs. Laser Robots*.
- STEM Outreach Programs** — Participated in REACT@TAMU and chemistry outreach initiatives for Katy ISD students.

Graduate Courses

Texas A&M University

College Station, TX, USA

Graduate Student, Department of Chemistry

2025 – Present

- **Quantum Mechanics (Physics Department)** — Schrödinger wave equation, bound states of simple systems, collision theory, representation and expansion theory, matrix formulation, perturbation theory.
- **Modern Computational Physics** — Modern computational methods with emphasis on simulation such as molecular dynamics and Monte Carlo; applications to condensed matter and nuclear many-body physics and to lattice gauge theories.
- **Principles of Quantum Mechanics (Chemistry Department)** — Classical mechanics and development of wave mechanics; application of wave mechanics to special chemical problems.
- **Statistical Thermodynamics** — Methods of statistical mechanics based primarily on Boltzmann statistics; approach to thermodynamics through partition functions.
- **Frontiers in Chemical Research** — Present status of research in a variety of significant chemical fields.
- **Ethics of Chemical Research** — Ethical issues in chemical research and scholarship and methods for resolution of such issues; includes Texas A&M University Policies and Procedures, ethics and scientific truth, ethics and other scientists and ethics and society; case studies.
- **Methods of Teaching Chemistry Labs** — An introduction to teaching methods associated with the teaching of introductory chemistry laboratories using graduate teaching assistants.

Skills

Programming	Python, C++, Fortran, MATLAB
Quantum Chemistry	Psi4, PySCF, ORCA, Gaussian, MCTDH-Heidelberg
Software	Scientific code development, performance optimization, modular design, API development
Computing Tools	Linux, Bash/Zsh, Git, Jupyter, LaTeX
Visualization	VMD, Avogadro, Chemcraft
Languages	Persian (native), English (IELTS 8.0), French (DALF C2), Latin (proficient), Spanish (elementary)
Communication	Scientific communication, technical writing, presentations, teamwork, leadership, time management

References

Dr. Arkajit Mandal

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Station, TX

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Dr. Daniel Tabor

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Station, TX

Assistant Professor of Chemistry

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Dr. Dong-Hee Son

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Professor of Chemistry

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